SEDM: Selective Self-Evolving Distributed Memory for Agents

Yichen Liu, Anna Min, Yinzhi Wang et al. · arXiv 2509.09498v4 · Original diagram & charts

HarnessRSI 65 ★★

evidence: retained

Paper

1 What changes, what is kept

1 Episode → candidate memory
SCEC: input, output, tool summaries, seeds + configuration hashes.
Workers return statistics + versions.

2 Paired replay → keep or reject
Held-out: 3 similar admission tasks, 3 runs per arm; keep if reward Δ ≥ 0.
Filter failed sources / answer copies.

3 One shared memory, managed
Rank: relevance × utility; move tiers.
Merge duplicates; distill patterns; isolate harmful items; decay unused.

4 Retrieve → recipient-side gate
CMG predicts success: USE and SKIP.
Learn both outcomes, including ties; calibrate on a separate split.

5 Inject only if expected gain pays
USE if calibrated success gain exceeds memory-length cost penalty.
Otherwise SKIP the retrieved set.

Diagram: held-out protocol. Tables 1–3 use source-task admission, not rerun held-out.

Artifact: shared memory. CMG trains Qwen3-4B / LoRA; gpt-4o-mini is fixed.

Limit: no replica/sync protocol or network-cost benchmark. Evaluation freezes memory.

2 Biggest gain: open-domain F1

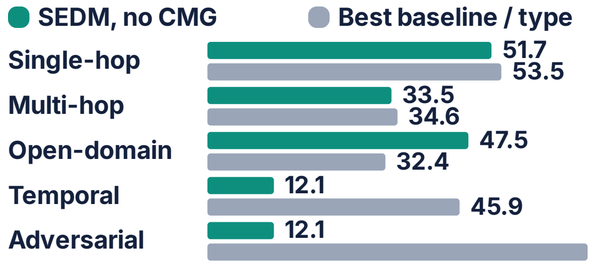


Table 1 · gpt-4o-mini · original admission · no CMG. Baselines: G-Memory for first three types, A-Mem temporal, LoCoMo adversarial. No variance reported; memory budgets may differ.

4 Where gains are absent or thin

- **LoCoMo:** open-domain leads; temporal 12.1 vs 45.9, adversarial 12.1 vs 69.2.
- **Transfer can fail:** LoCoMo memory → HotpotQA EM 34, equal to no memory. FEVER memory gives 41 vs in-domain 39 (Table 3).
- **Frozen controller memory:** always-use 0.518 vs no-memory 0.534; CMG 0.546. Only one test conversation; differences within its noise.
- **Table 5:** independent banks / anchors, not a paired controller ablation.

3 Controls, uncertainty and cost

Retention ablation · Table 2, no CMG

Method	FEVER	HotpotQA
G-Memory	62	38
+ admission	64	37
+ controller	66	39

Same memory budget, not full compute. No seeds/CIs or baseline tuning budgets reported. No diffusion ablation.

Controlled HotpotQA gate · Table 4

Policy	Success	Input tokens
No memory	0.249	1,629
Always use	0.333	3,736
CMG (val. λ = 0)	0.329	3,047

489 questions · seed 42 · F1 ≥ 0.5. CMG CI [0.288, 0.372] overlaps always-use [0.292, 0.376]: saves 18%, not an accuracy win.

Same paired runs / memory for gates; CoT baseline is zero-shot, not its trained checkpoint.

Held-out LoCoMo gate · Table 5

Admission-only: CMG 0.556 vs always 0.539 / skip 0.544; saves 19%. Three-seed mean, 193 questions; SBERT ≥ 0.7. No rowwise CIs.

Validation: checkpoint + λ; calibration split fits temperatures. HotpotQA val-selected: 0.318 ± 0.015 (Table 6).

Guards: split/hash and held-out answer-copy checks. HotpotQA uses gold evidence; related facts may overlap.

Cost: paired data ≈ $4.5; 24.7M input / 1.3M output. Full build, replay, merge, sync and GPU costs not itemized.

Takeaway

Test what enters memory; separately decide whether to use it. The held-out evidence supports a quality–context-cost trade-off over frozen memories, not continuously self-improving distributed agents.

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Reports: ablations, transfer, gate seeds, split guards.
Missing: full compute, online evolution, build/sync cost.